MASHUK ELAHI

Critical Facilities & Commissioning Engineer | MV 11 kV, UPS, EPMS & BESS

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PROFESSIONAL SUMMARY

Electrical and commissioning engineer with 16 years energising and handing over MV/LV power systems across semiconductor, marine, offshore-wind, and industrial plants. Directed critical-power hook-up and energisation across 30 MW of N+1 semiconductor capacity, closing out with zero audit rework and first-pass PLC/SCADA interlock validation against 99.99% uptime (Tier III-equivalent) targets. Deep hands-on scope spans MV 11 kV switchgear, UPS and BESS integration, protection and EPMS/SCADA, offshore-wind substations, vessel newbuilds, and IECEx/ATEX hazardous-area design. CDCP certified; ISO 9001 and ISO 50001 Lead Auditor. Seeking senior commissioning and critical-facilities roles in Singapore, Europe, and the UAE.

CORE COMPETENCIES

- **Power systems:** MV/LV distribution; MV 11 kV switchgear; transformers, protection relays, and grounding; UPS, standby generators, and BESS integration
- **Controls & automation:** EPMS/SCADA, BMS/BAS, and PLC/SCADA; ABB 800xA; IEC 61850; engineering workflow automation with Python and AutoCAD Electrical
- **Commissioning & reliability:** FAT/SAT, point-to-point checks, energisation, functional performance and integrated systems testing; N+1, 2N/2N+1 redundancy; Uptime Institute Tier topologies; 99.99%+ uptime SLAs and concurrent maintainability; MOP/SOP/EOP; critical cooling interfaces (chiller plant, CRAH/CRAC, PCW)
- **Energy, hazardous areas & risk:** PUE tracking and energy-performance monitoring (ISO 50001); cogeneration and autonomous renewables; IECEx/ATEX design; marine electrical practice to IEC 60092-201; compliance auditing and gap analysis; technical bid evaluation and FMEA

PROFESSIONAL EXPERIENCE

Electrical Engineer | Hybrid & Critical-Power Systems

Strategic Marine Pte. Ltd., Singapore | Jul 2024 – Present

- Directed LV distribution design for a three-generator 450 ekW prime-power system — single-line diagrams (SLDs), load lists, and cable schedules — winning first-pass approval from four independent certification bodies with no rework.
- Engineered aluminium-hull IT earthing architecture with single-point common-zero bonding in strict adherence to IEC 60092-201, and executed stray-current rectification on fast crew boats, severing stray-current loops by upgrading isolation switches to DPST.
- Integrated a 72 kWh lithium-ion BESS, battery management system (BMS), and DC/DC inverters into a hybrid power architecture that cut fuel and energy draw by up to 50% at low load; drove the BMS-to-EPMS interface and FAT through to handover.
- Tracked and managed electrical-system weights exceeding 7,800 kg across load lists and cable schedules, feeding comprehensive vessel stability calculations.
- Automated SLD generation and VDR alarm-list production using Python, AI agents, and AutoCAD Electrical scripting, and conducted technical gap analyses of navigation and communication equipment against class and contract specifications.

Senior Project Engineer | Semiconductor Critical-Power Commissioning

Exyte, Singapore | Feb 2023 – Jul 2024

- Spearheaded 300 mm cleanroom critical-power hook-up, energisation, and functional performance testing on 30 MW (33 MVA) of critical load at 1.5 kW/m² fab power density — zero audit rework and first-pass PLC/SCADA interlock validation against 100% tool-uptime and N+1 objectives (99.99% uptime SLA equivalent).
- Introduced prefabricated power skids that cut on-site installation labour by 50% and removed contamination risk during live energisation, coordinating with chilled-water and process-cooling-water (PCW) commissioning teams.

E&I Engineer | Industrial Power, Controls & Hazardous Areas

Phoenix Mecano S.E.A. Pte. Ltd., Singapore | Jan 2021 – Jan 2023

- Directed the complete electrical and instrumentation (E&I) scope for an onshore 10 MW cogeneration plant: transformers, MV/LV switchgear, a 400 kVA UPS, SCADA integration, tie-in design, instrument loops, and technical bid evaluation.
- Delivered Malaysia's first solar-wind autonomous remote-power installation on a 24 V DC solar-battery bus, removing routine diesel-generator use and enabling unmanned site operation.
- Designed IECEx/ATEX hazardous-area systems, including explosion-protected (Ex) equipment selection, intrinsically safe barriers, and zoning studies.

E&I Superintendent | Offshore Power Systems & Commissioning

Keppel FELS / Keppel Offshore & Marine, Singapore | Nov 2012 – Dec 2020

- Led commissioning of MV 11 kV switchgear and ABB 800xA fire-and-gas systems, holding MV/LV power reliability through final handover with crews of up to 100 people.
- Energised an offshore-wind substation to full IEC 61850 protection-relay conformity with zero punch-list items at handover.
- Authored 30 E&I risk assessments and mentored 15 engineers through Innovation Quality Circles.

Assistant Electrical Engineer | Vessel Newbuild Electrical

Ananda Shipyard & Slipways Ltd., Bangladesh | Sep 2009 – Oct 2012

- Facilitated electrical installation, testing, and commissioning on vessel newbuilds, collaborating with senior engineers on equipment energisation and documentation.

EDUCATION

- Master's Degree in Continuing Education in Sustainable Electricity Generation (UCAM, 60 ECTS, título propio), UCAM Universidad Católica de Murcia, Spain — Apr 2026
- MSc Digital Transformation, London Metropolitan University, UK — Mar 2026
- Master of Business Administration (MBA), University of Sunderland, UK — Jun 2025
- BSc Electrical and Electronic Engineering, Bangladesh University — Jul 2008
- Diploma in Electronics Engineering, Jessore Polytechnic Institute — Jan 2004

CERTIFICATIONS

- CDCP — Certified Data Centre Professional, EPI — 2026
- ISO 50001:2018 Energy Management Lead Auditor, GAQM / IGC
- ISO 9001:2015 Quality Management Lead Auditor, GAQM / IGC
- IECEx Ex 001: Basic Principles of Protection in Explosive Atmospheres — valid through Oct 2027
- ABB Power System / LV Switchboard Training
- Fluke DSX2-8000 Cable Analyzer / Copper Certifier Training

TECHNICAL TOOLS

ABB 800xA; PLC/SCADA systems; IEC 61850 protection relays; EPMS/SCADA; AutoCAD Electrical; Python (engineering workflow automation); Fluke DSX2-8000 Cable Analyzer / Copper Certifier.